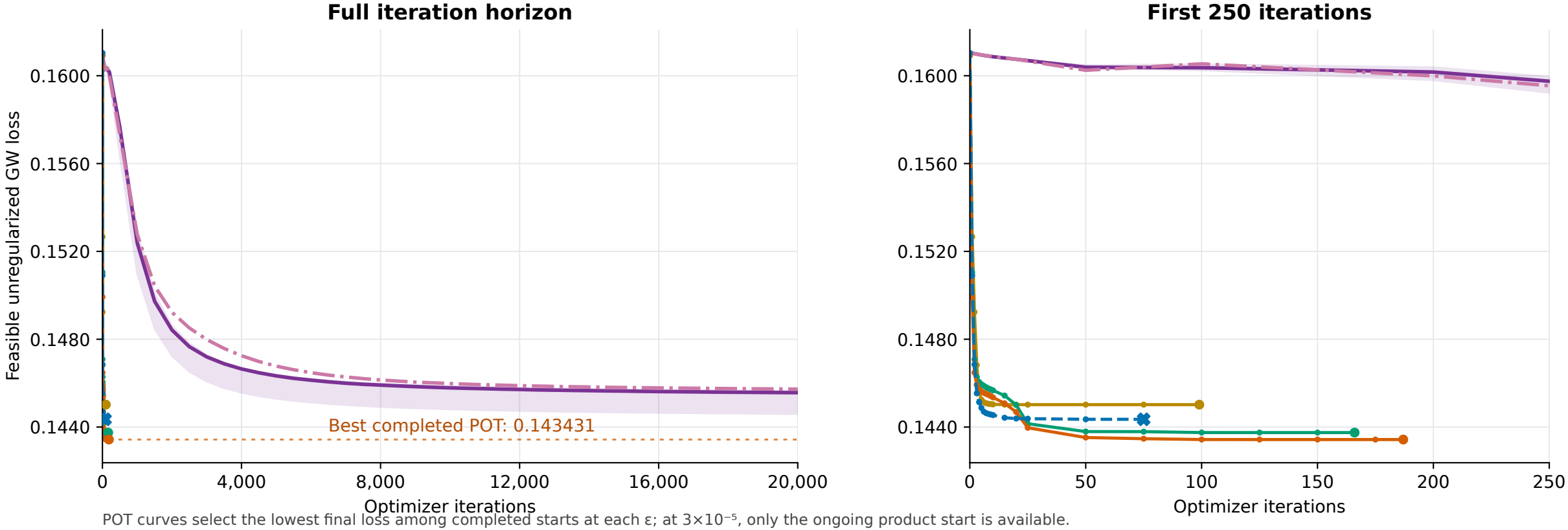
# BMMC · 6,224 cells: FRAMES and the POT entropy sweep

FP64 · Feasible unregularized GW loss · Lower is better · Saved results as of 21 Oct, 09:58 Paris

FRAMES, 1,024 pairs: median of 3 seeds (band: range)
FRAMES, 4,096 pairs: CPU, seed 811
POT, ε=10⁻³: permutation mixture, seed 903
POT, ε=3×10⁻⁴: permutation mixture, seed 903
POT, ε=10⁻⁴: permutation mixture, seed 903
POT, ε=3×10⁻⁵: product start (unfinished)




POT curves select the lowest final loss among completed starts at each ε; at 3×10⁻⁵, only the ongoing product start is available.

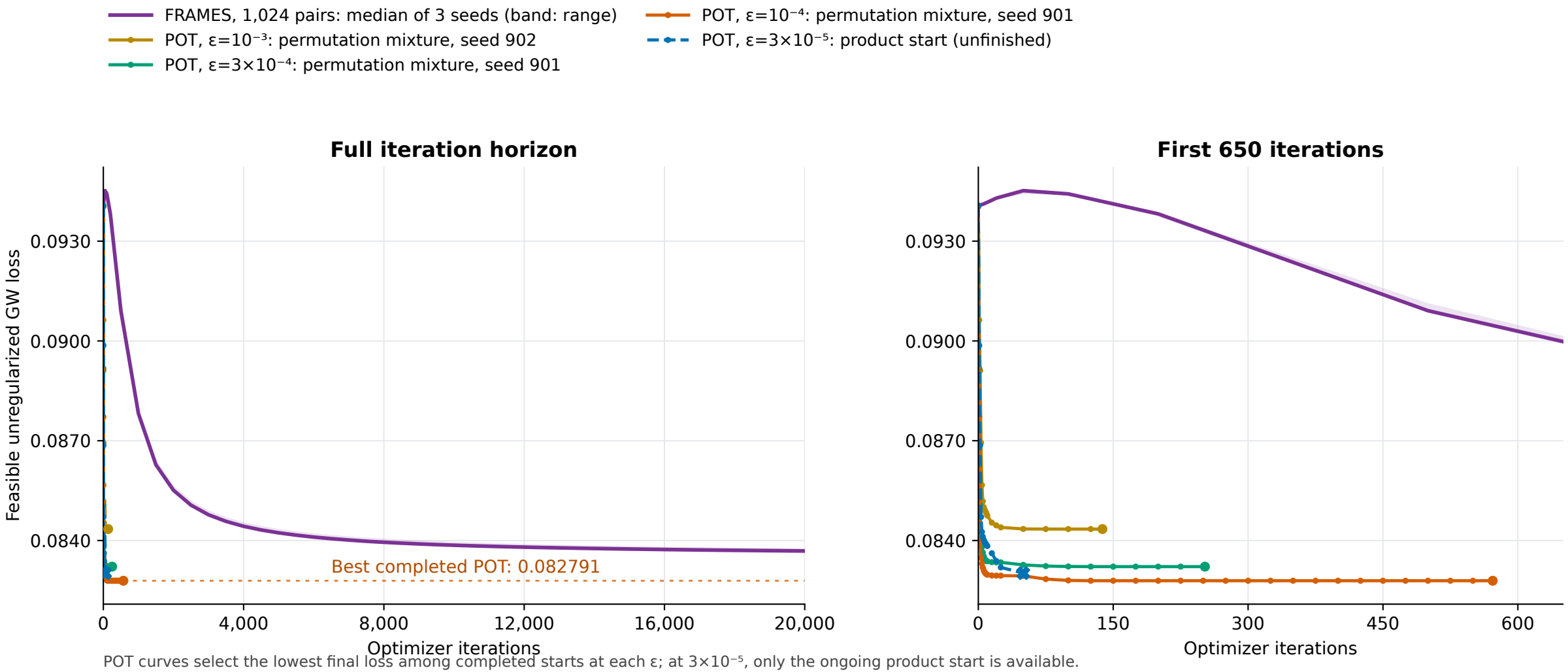
POT dots are measured losses; × marks an unfinished trajectory. The horizontal dotted line is a final-value reference only.

FRAMES uses product initialization; POT initialization varies. Iterations have different computational costs.

The 4,096-pair CPU run is one replicate; CPU/CUDA random streams differ, so batch size is not isolated.

# SNARE-seq · 10,000 cells: FRAMES and the POT entropy sweep

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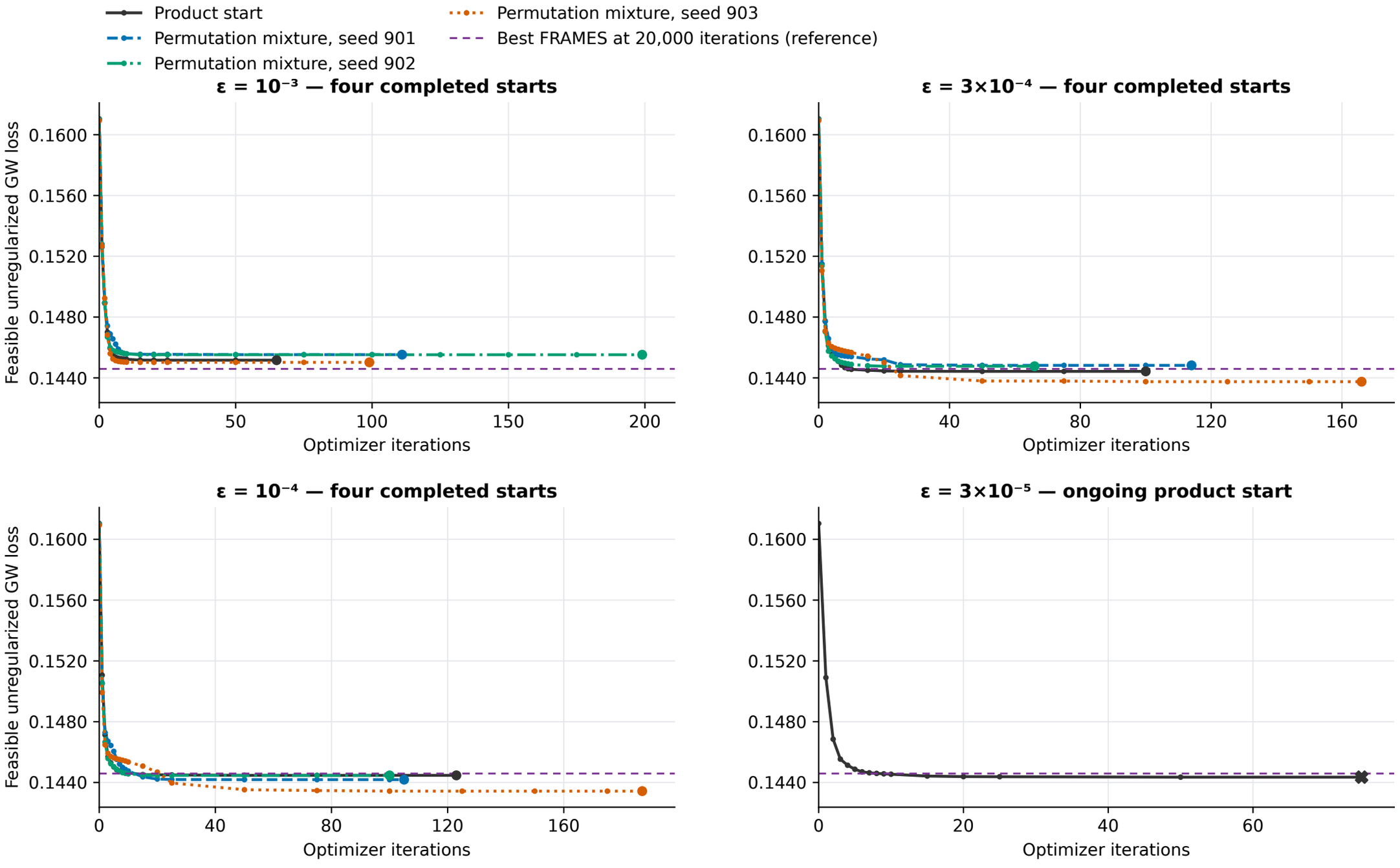

POT curves select the lowest final loss among completed starts at each ε; at $3\times10^{-5}$, only the ongoing product start is available.

POT dots are measured losses; × marks an unfinished trajectory. The horizontal dotted line is a final-value reference only.

FRAMES uses product initialization; POT initialization varies. Iterations have different computational costs.

# BMMC · 6,224 cells: every available POT trajectory

Full-gradient entropy-regularized GW updates · Sinkhorn tolerance $10^{-7}$ · 21 Oct, 09:58 Paris

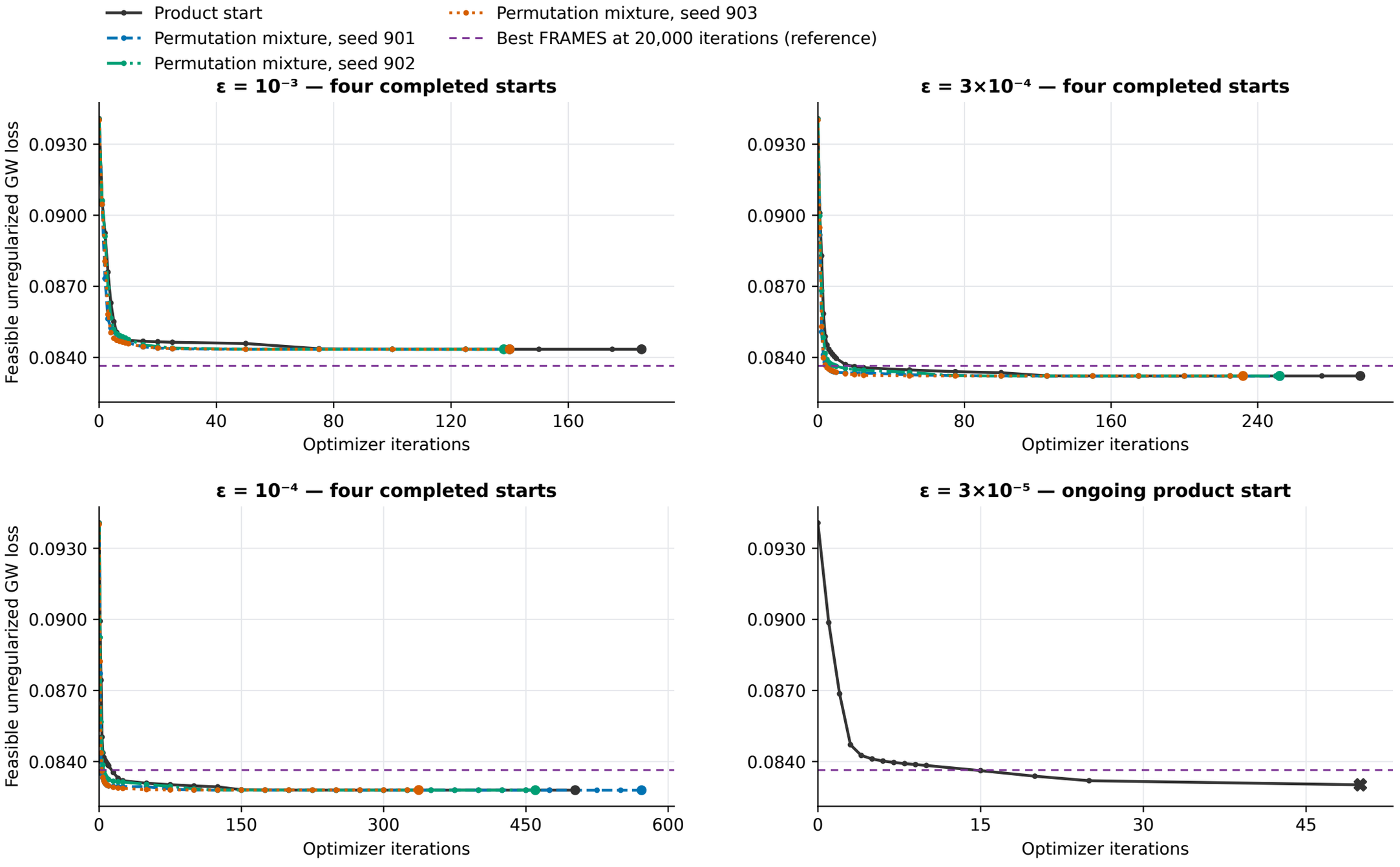



Dots are recorded diagnostics; × marks the last diagnostic of an unfinished run. All panels use the same loss scale.

Other starts at ε=3×10⁻⁵ and all ε=10⁻⁶ runs have no trajectories yet. Dashed purple lines are references, not POT runs.

# SNARE-seq · 10,000 cells: every available POT trajectory

Full-gradient entropy-regularized GW updates · Sinkhorn tolerance $10^{-7}$ · 21 Oct, 09:58 Paris




Dots are recorded diagnostics; × marks the last diagnostic of an unfinished run. All panels use the same loss scale.

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